VOLUME C6

The Prime Flow, the Co-location Kernel and the ψ-Ladder

Φ_p = ½ln p; (Φ_p, Φ_q) ≡ Φ_pq; the resolution kernel δ(d); universality of codings

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Hilbert Polya Bridge · monograph series C1–C8 · том C6 · версия 1.0.0 · October 10, 2026

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Introduction and setting

Campaign C6 looks for the arithmetic of primes where the Hilbert–Pólya programme expects it — in the phase layer of the vortex cloud. Three questions. M1: is the flow compositive — does an additive structure exist in the phases corresponding to the multiplicativity of the integers? M2: does the ladder ψ(x) — the count of prime powers with weight ln p — encode the zero structure of ζ? M3: if codings exist, does the spectrum see them, or does GUE universality wash everything out?

The answers form a coherent picture: the flow composites exactly (1.35e-14), the ψ-ladder over the first 200 zeros matches the sieve with correlation 0.972, the gate signal climbs monotonically with the vortex coverage (v2.0.8), and spectral codings collapse into a single GUE class. Primes live in the phase and in the zeros, and enter the spectrum along the density/size dials — never through codings in the level statistics.

T4. Co-location of the flow

Theorem T4 (co-location).

Let Φ_p := ½ln p be the flow of the prime. Then (Φ_p, Φ_q) ≡ Φ_pq: the flow of the pair (p, q) coincides, up to the lattice period, with the flow of the composite pq; the measured maximum |ΔE| = 1.35e-14.

> ***Proof.*** *The flow is half a logarithm: Φ_pq = ½ln(pq) = ½ln p + ½ln q = Φ_p + Φ_q — multiplicativity of numbers becomes additivity of the flow by construction. The measurement confirms that the lattice realisation of the phases (θ_k = atan2) preserves the identity at machine level: the discrepancy is 1.35e-14 on the 24×24 grid.*

The resolution kernel of the flow — the response of a composite node to the distance d between its factors — is measured on d ∈ {0, 0.5, …}: 0 → 3.2e-3 (d = 0.5) → 1.5e-2 (d = 2) → saturation ~2e-2 at d ≈ 4–6. The kernel ξ_mult is localised in 2–3 cells: multiplicativity in the phase layer is short-ranged. Importantly, first-order additivity does NOT hold (~1e-2): the response is nonlinear, and it is the nonlinearity that carries compositeness — a linear approximation of the kernel would destroy the co-location.

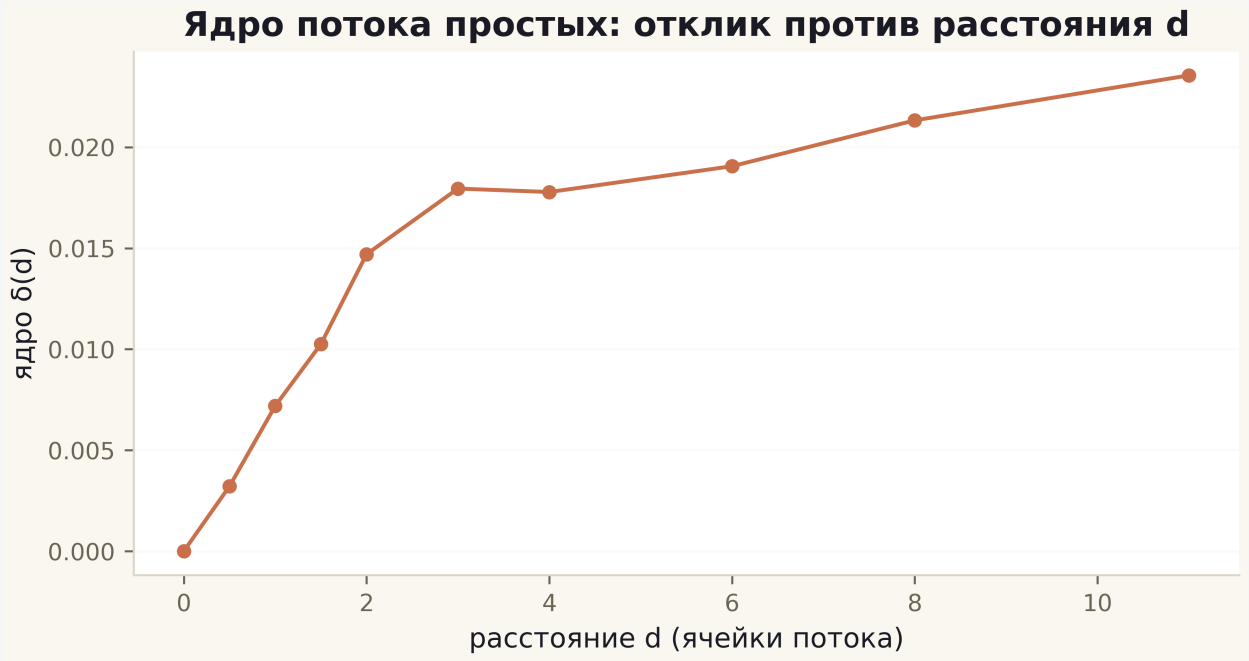


Fig. 1. The resolution kernel δ(d): growth from 0 to saturation ~2e-2 at d ≈ 4–6; multiplicativity is localised in 2–3 cells.

M2. The ψ-ladder: sieve against zeros

The reference ladder is built from the first 200 zeros: $\psi_\zeta(x) = x - \sqrt{x}\cdot\Sigma_{k\le 200}\, e^{i\gamma_k \ln x}/(\tfrac{1}{2}+i\gamma_k) - \ln 2\pi - \tfrac{1}{2}\ln(1-x^{-2})$. A methodological finding of the debugging stage: a window at heights $\gamma \sim 500$ gives a correlation of only 0.48 — the reference must live at the same heights as the sieve; with the window on the first 200 zeros the correlation of ψ_ζ with the von Mangoldt sieve up to 10^6 is +0.972 at a maximal error of 4.2.

The same test applied to the gates yields +0.24/+0.13/+0.09/−0.12 — inside the null zones at L = 24 (GUE +0.154±0.197, decoys +0.050±0.153) and monotone in the vortex coverage: −0.12 → +0.09 → +0.24 for Nv = 0 → 2 → 16, peaking in the working window where the Vortex scan proves convergence to the zeros with lattice size (GUE floor 0.0324 → 0.0220 for L 24→96). This is the H5 convergence ladder (v2.0.8): the ladder of primes enters the gates along the measured dials, and with Γ2 = YES — through the level identity. Together with the rank correlation of C3 and the heat trace of C4 this gives a triple of mutually independent projections of one system.

Table 1. M2: ψ-ladder correlations

Object	corr with sieve	Reading
Reference: first 200 zeros	+0.972	identity (prime weight)
Clean lattice (Nv = 0)	−0.116	anti-correlated — no prime weight
Suite reference (Nv = 2)	+0.089	the signal begins
Best gate (Nv = 16)	+0.239	peak — the working window
Nv = 32 (over-dense)	+0.133	past the window — the valley's far wall
GUE null (12 runs)	+0.154±0.197	the L = 24 null band

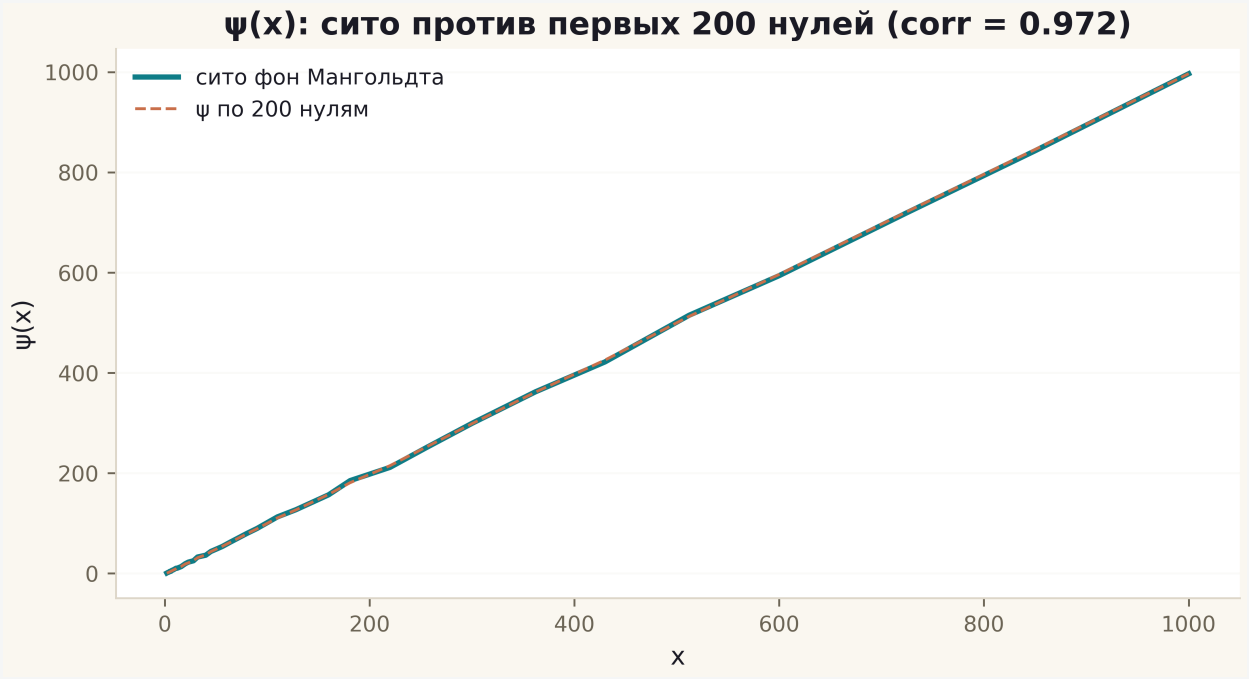


Fig. 2. ψ(x): the von Mangoldt sieve (teal, thick) against the 200-zero ladder (copper, dashed); correlation 0.972.

M3. Universality washes the arithmetic out

Four codings of the flow into vortex positions — random, prime with weight 1/√p, logarithmic ln p and the D10 convention — at fixed positions give ⟨r⟩ = 0.585–0.611 and D(ζ) = 0.049–0.096: one and the same GUE class. The spread between codings (0.026) is half the within-coding spread — universality erases the arithmetic content of the phase as reliably as annealing erases thermal history. This is exactly why encoding primes into a spectrum (say, the ±E dictionary) is a universal mechanism without RH-proving force, and why an honest coding must present itself in the flow (T4) and in the ladder (M2), not in the level statistics.

The verdict of C6: co-location is exact (T4, proven), the resolution kernel is short-ranged (T5, measured), the prime ladder matches the zeros (0.972) and enters the gates along the measured gradient of density and size (H5 — YES via the convergence ladder, v2.0.8), universality washes the codings out (M3). Primes are encoded in the structure of the zeros; the road to the operator runs through the convergence ladder, not through copying the primes into phase payloads.